

# Operations Management

## Session 1: Introduction to Operations Management

### Outline:

- Course Organization
- Overview and Goals
- Operational Capabilities
- Fundamentals of Business Processes
  - Little's Law
  - Key Process Metrics: Throughput, Inventory, Cycle Time
  - Cash-to-Cash Cycle
  - Circular Flows

# The Ops Team



## **René Caldentey**

Booth Faculty Operations Management Group

Former Faculty NYU Stern

Research Interests: Revenue Management, Retailing, Queueing Theory



## **Tong Xie**

Booth PhD Student: Operations Management

Research Interests: Machine Learning, AI, Algorithm Collusion



## **Maximiliano Stock**

Booth PhD Student: Operations Management

Research Interests: Inventory Management, Supply Chains, Logistics



BOB – Booth Ops Bot

Role: AI Teaching Assistant | 40801 Operations Management

# Course Organization

## GRADING COMPONENTS

- **Individual Activities 60%**
  - Class Participation 10%
  - Takeaways 5% (Week 1 only)
  - Midterm Exam 15% (transferable score)
  - Final Exam 30%
- **Team-Based Activities 40%**
  - Group Assignments 20%
  - Group Project 20%

## FINAL GROUP PROJECT: Course Takeaways

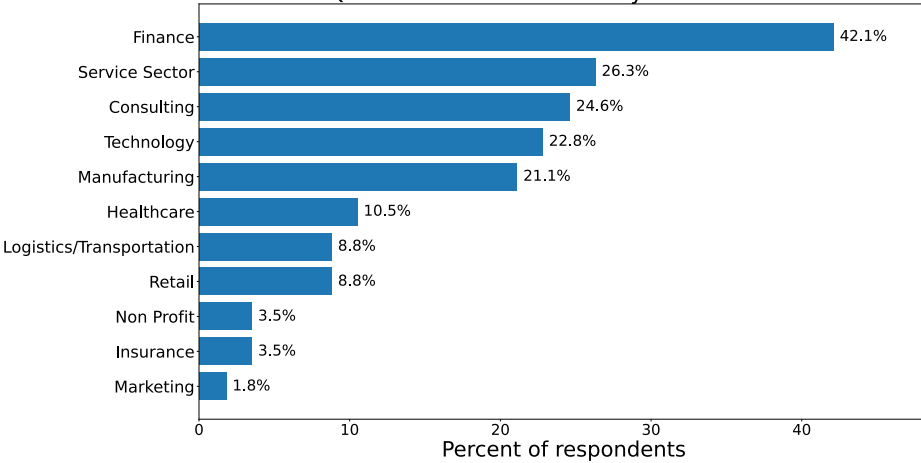
- Individual Takeaways (five by team member)
- Final Report: Develop two takeaways
- Final Presentation: Choose one.

## REVIEW SESSIONS (4):

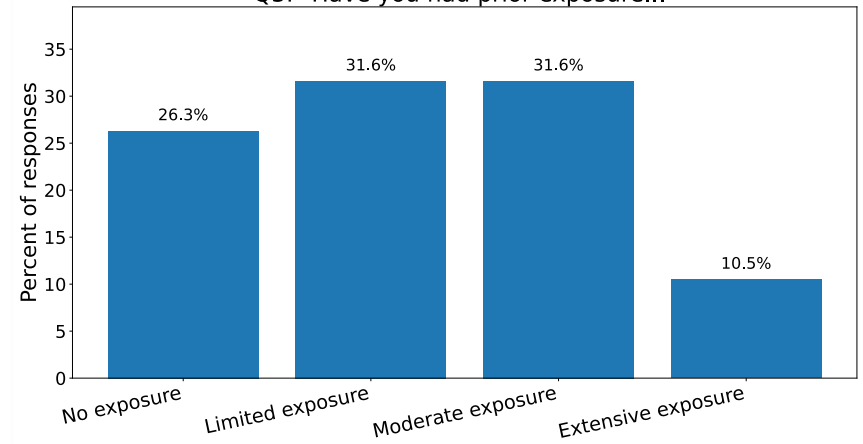
- Two per week
- Wednesday & Friday (after class)

# Pre-Course Survey

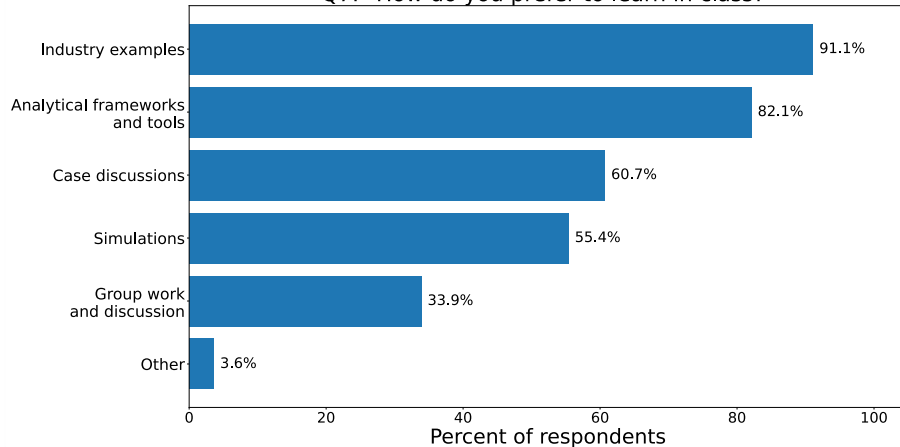
Q2: Which industries have you worked in?



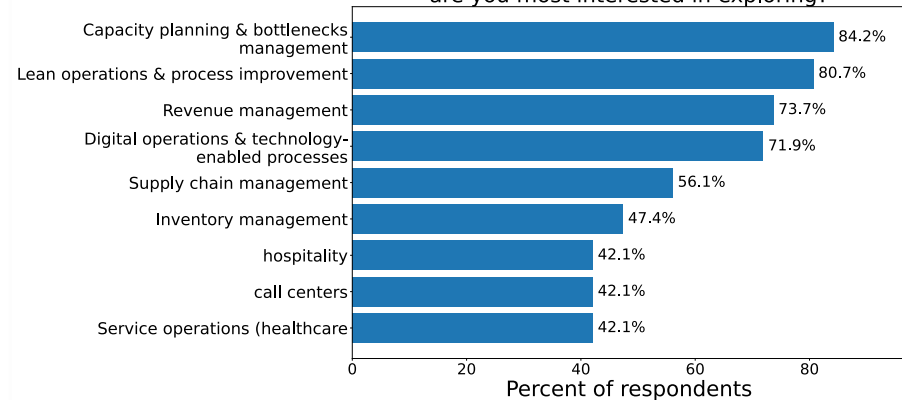
Q3: "Have you had prior exposure..."



Q7: "How do you prefer to learn in class?"



Q8: Which topics in Operations Management are you most interested in exploring?



# Pre-Course Survey



# Pre-Course Survey

*“...how AI improve ops management.”*

*“How to maximize the AI integration to help in supply chain, capacity planning and inventory ...”*

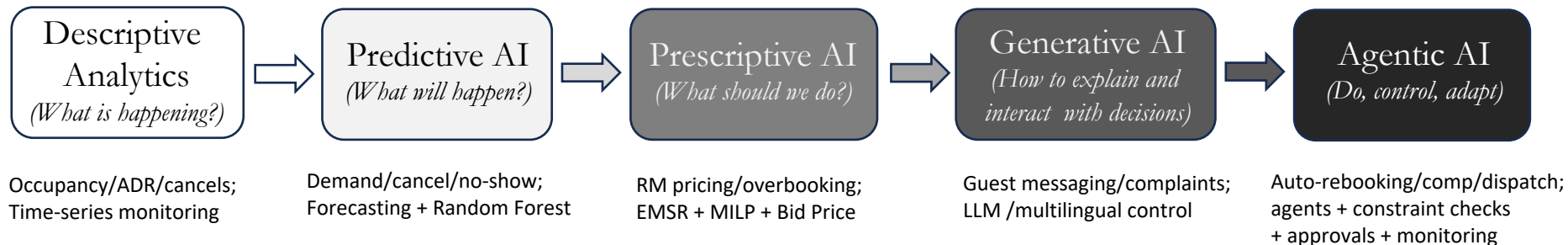
*“What is the impact of AI.”*

*“Operations impact from AI technology.”*

*“... and operations impact from AI technology.”*

*“I’m interested in how better data and analytics can improve operational decision-making.”*

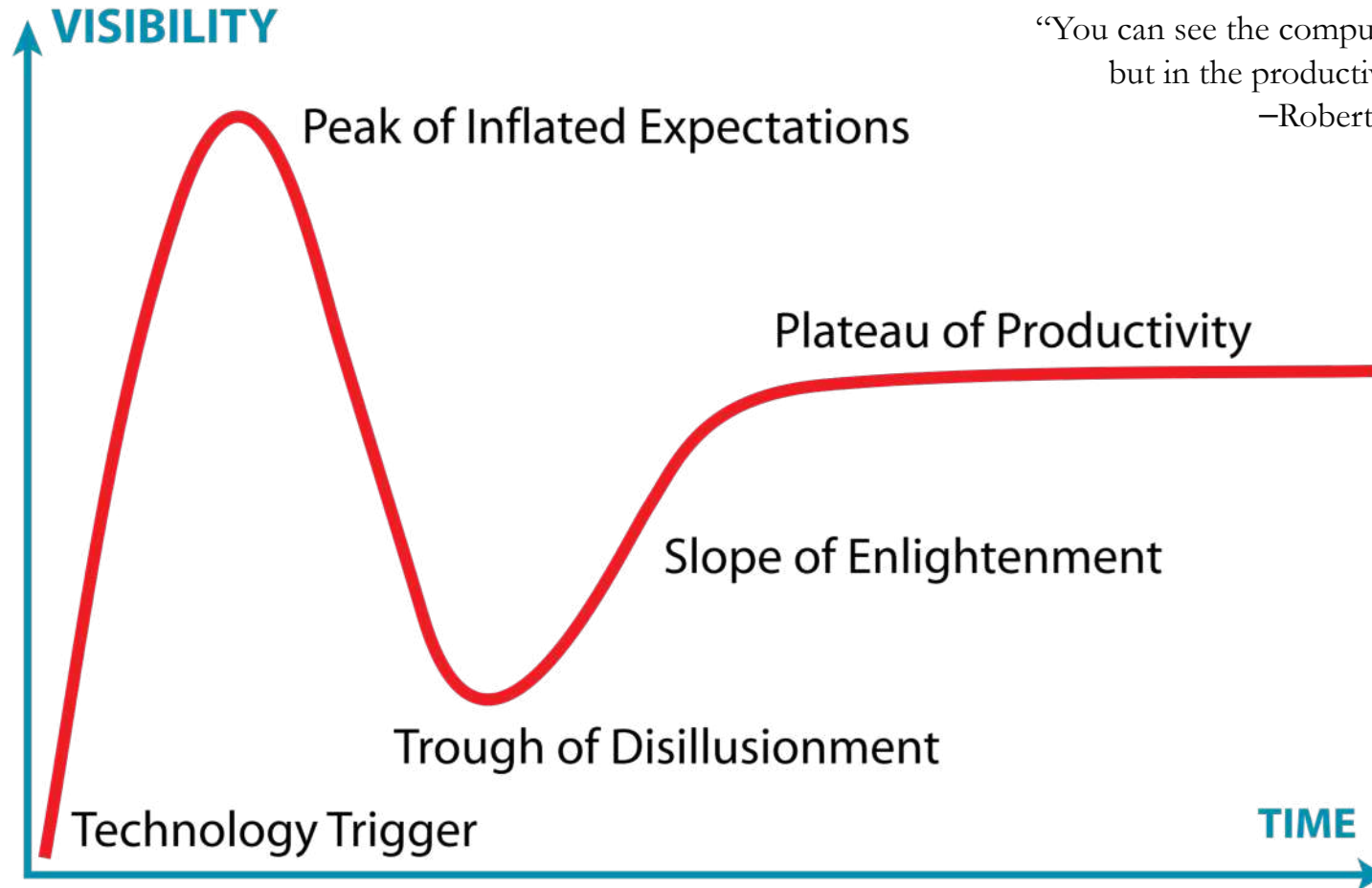
*“How can advanced analytics be used in real operational settings, not just theoretically?”*



# The Hype Cycle



“You can see the computer age everywhere  
but in the productivity statistics”  
—Robert Solow (1924-2023)



# Mixed Feelings

## How Americans' feelings on AI have changed since December

Which of the following describes how you feel about advances in artificial intelligence (AI)? Select all that apply. (% of U.S. adult citizens)

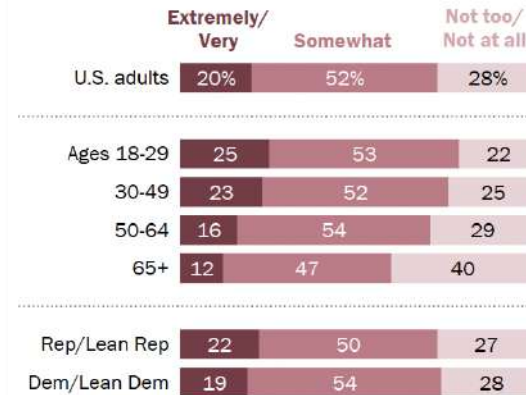


Note: Responses of "not sure" and "none of the above" are not shown.

YouGov

## 1 in 5 Americans who have seen AI summaries in search results find them extremely or very useful

Among U.S. adults who have seen AI summaries in search results, % who say they find them \_\_\_ useful



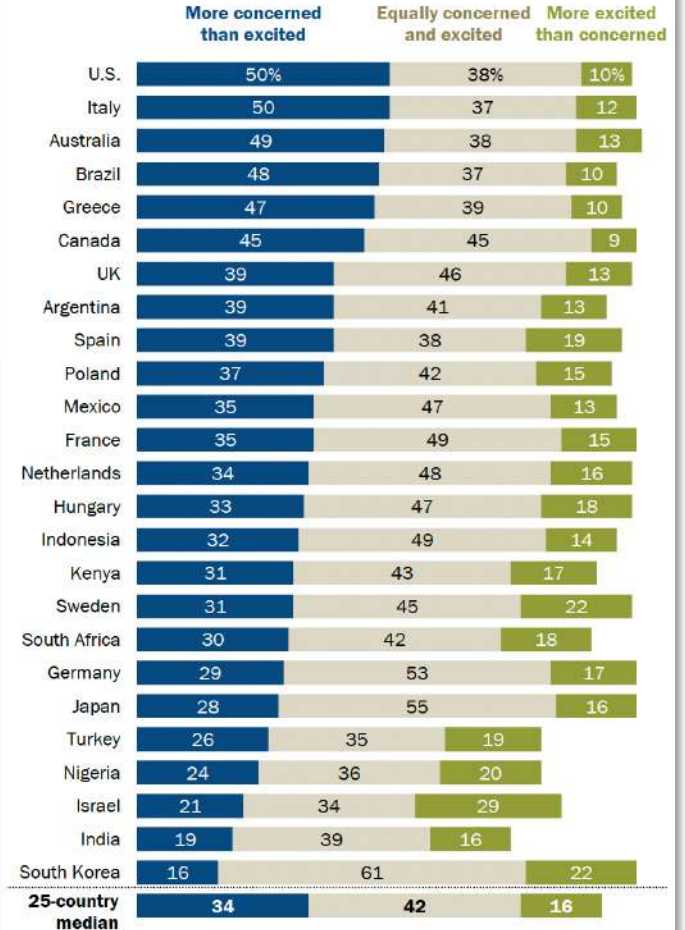
Note: Based on U.S. adults who have come across AI summaries in search results at least rarely. Respondents who did not answer are not shown.

Source: Survey of U.S. adults conducted Aug. 18-24, 2025.

PEW RESEARCH CENTER

## How do people around the world feel about the rise of AI in daily life?

% who say the increased use of artificial intelligence in daily life makes them feel ...



Note: Those who did not answer are not shown.

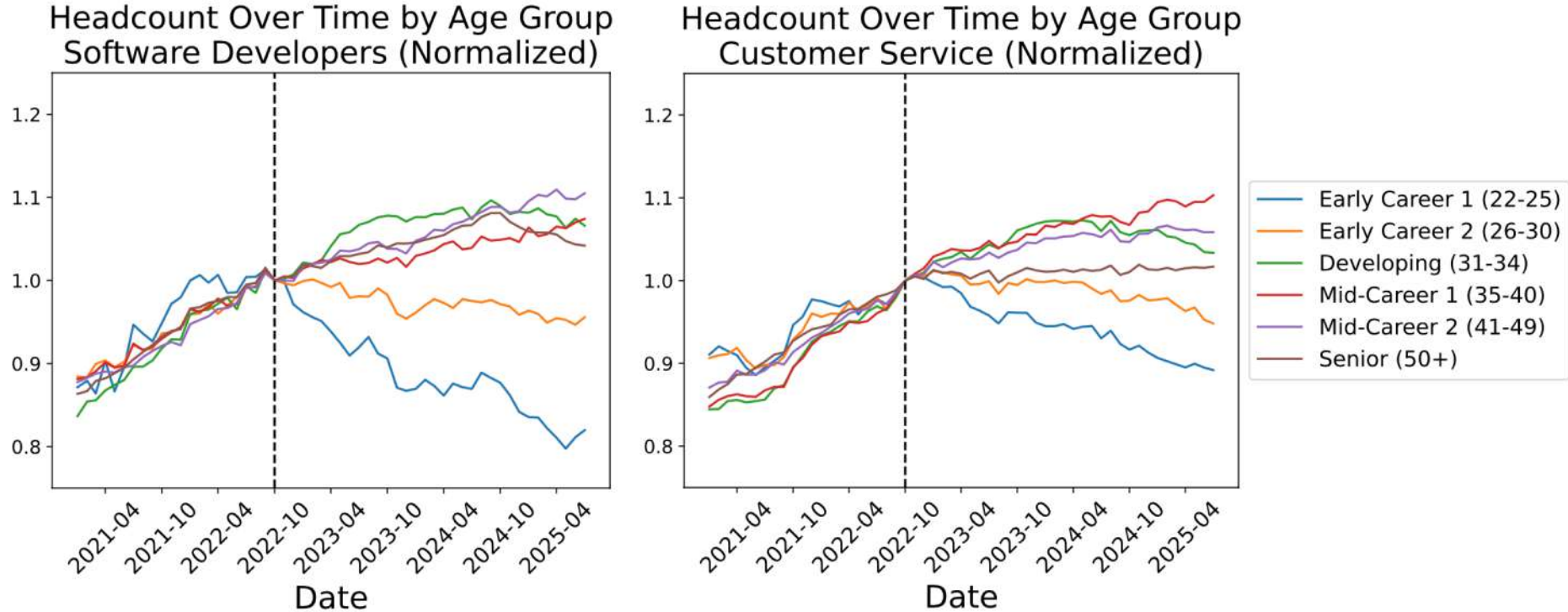
Source: Spring 2025 Global Attitudes Survey.

"How People Around the World View AI"

PEW RESEARCH CENTER



# Canaries in the Coal Mine?



Source: Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence. Brynjolfsson, Chandar, Chen (2025)

# Operations are how strategies become reality...

**Operations Management** is the collection of business activities involved in the design and management of the **processes** and **systems** that **transform** inputs into finished goods and services.

Processes and operating systems are everywhere in an organization and Operations typically employ the greatest number of employees and manage the largest investments in physical assets.

# When Operations Become the Bottleneck



# Goals for the course

## 1. How to “think” operationally?

**Why does improving one step make the system worse?** Local optimization vs. system performance.

## 2. How to structure and manage a business process so that operational capabilities are best achieved?

**E-commerce fulfillment:** How should picking, packing, and shipping be organized to deliver *fast and reliable delivery* at *low cost*?

## 3. How to “audit” (diagnose and improve) business processes using an operational lens?

**Service process reliability review:** Compare promised vs. actual delivery times → identify sources of variability → redesign policies to improve reliability.



# How to “think” Operationally?



# How to “think” Operationally?



London Tube Escalators During Morning Rush Hour ([Available in YouTube](#))



# To Walk, or Not To Walk

The  
Guardian



In a peak day, close to 5 million passengers use the London Underground system ([link](#))

# Operational Capabilities: The Four Views

## Capabilities



=

## Resources



+

## Processes



+

## Operational Culture



- Cost/Productivity
- Speed/Timeliness
- Quality
- Flexibility/Agility
- Expertise
- Innovation

- People/Skills
- Machinery & Equipment
- Financial Assets
- Facilities & Infrastructure
- IT Systems
- Data, AI Systems and Models
- Intellectual Property

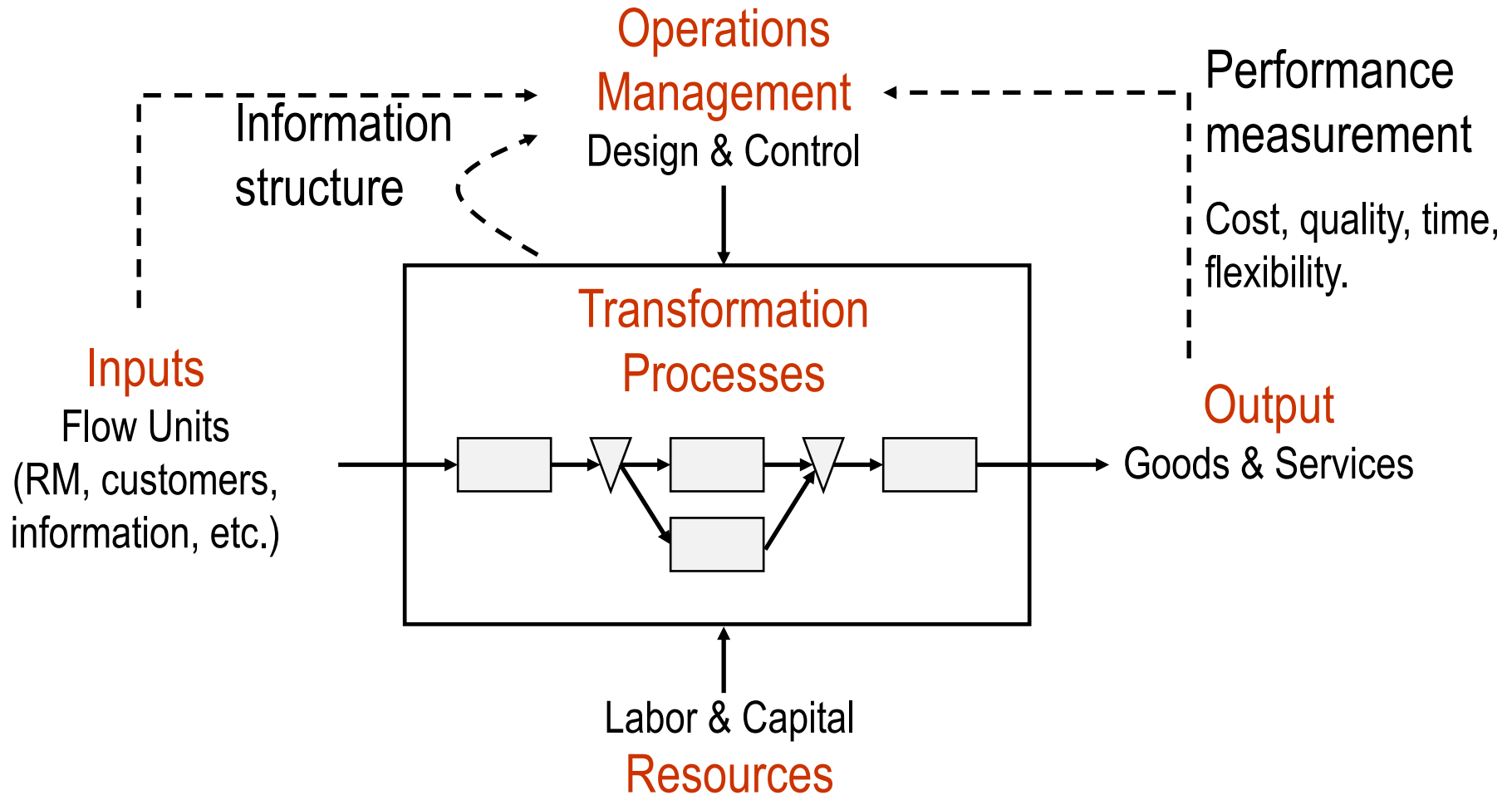
- Push/Pull
- Just-in-Time
- Lean Management
- Standard vs Customization
- Process Reengineering

- Focused
- Conservative
- Customer-Centric
- Efficiency-Driven
- Data-Driven
- Quality-Centric
- Compliance-Oriented
- Employee Empowered

CAPABILITIES = F(RESOURCES, PROCESSES, OPERATIONS CULTURE)



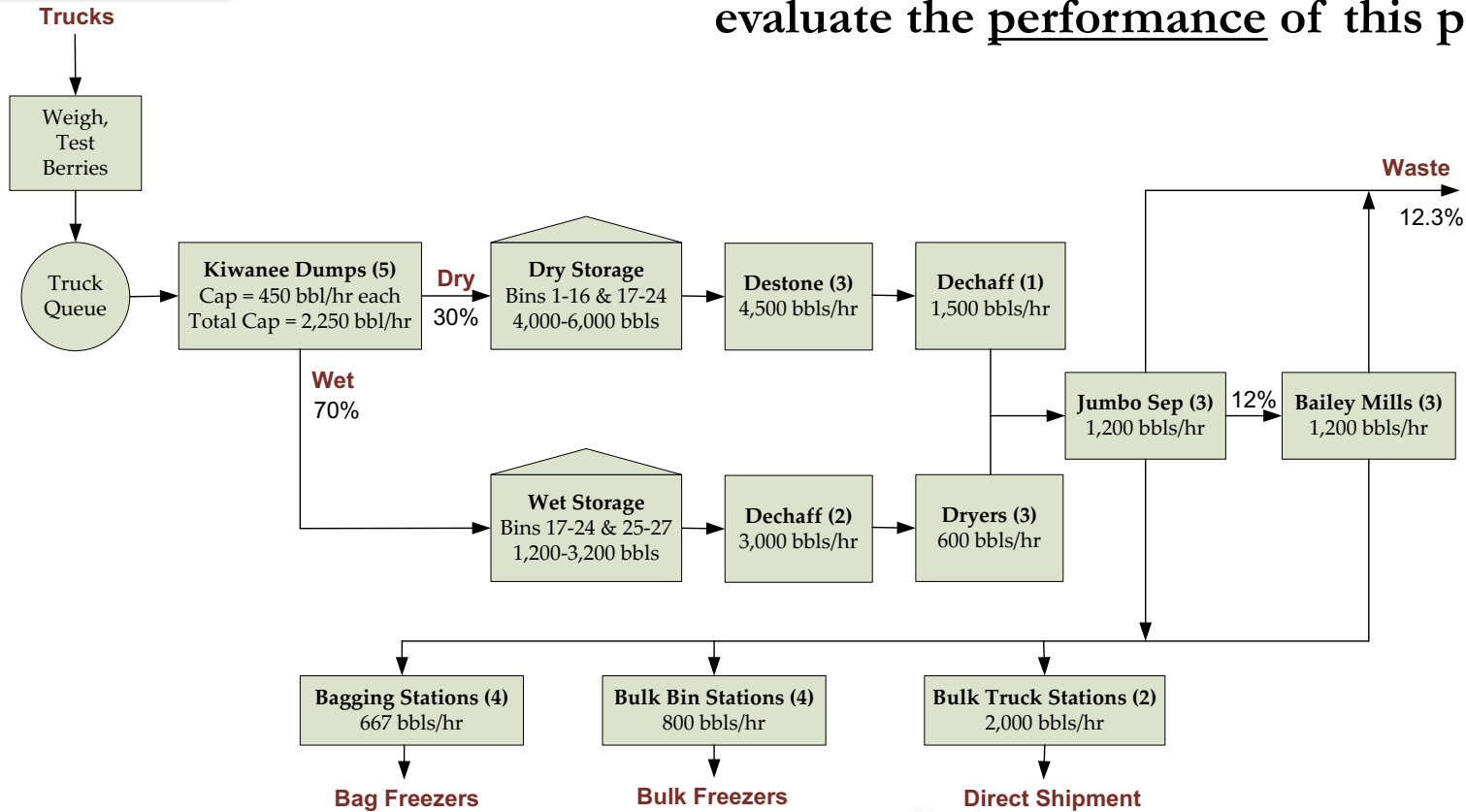
# Process View



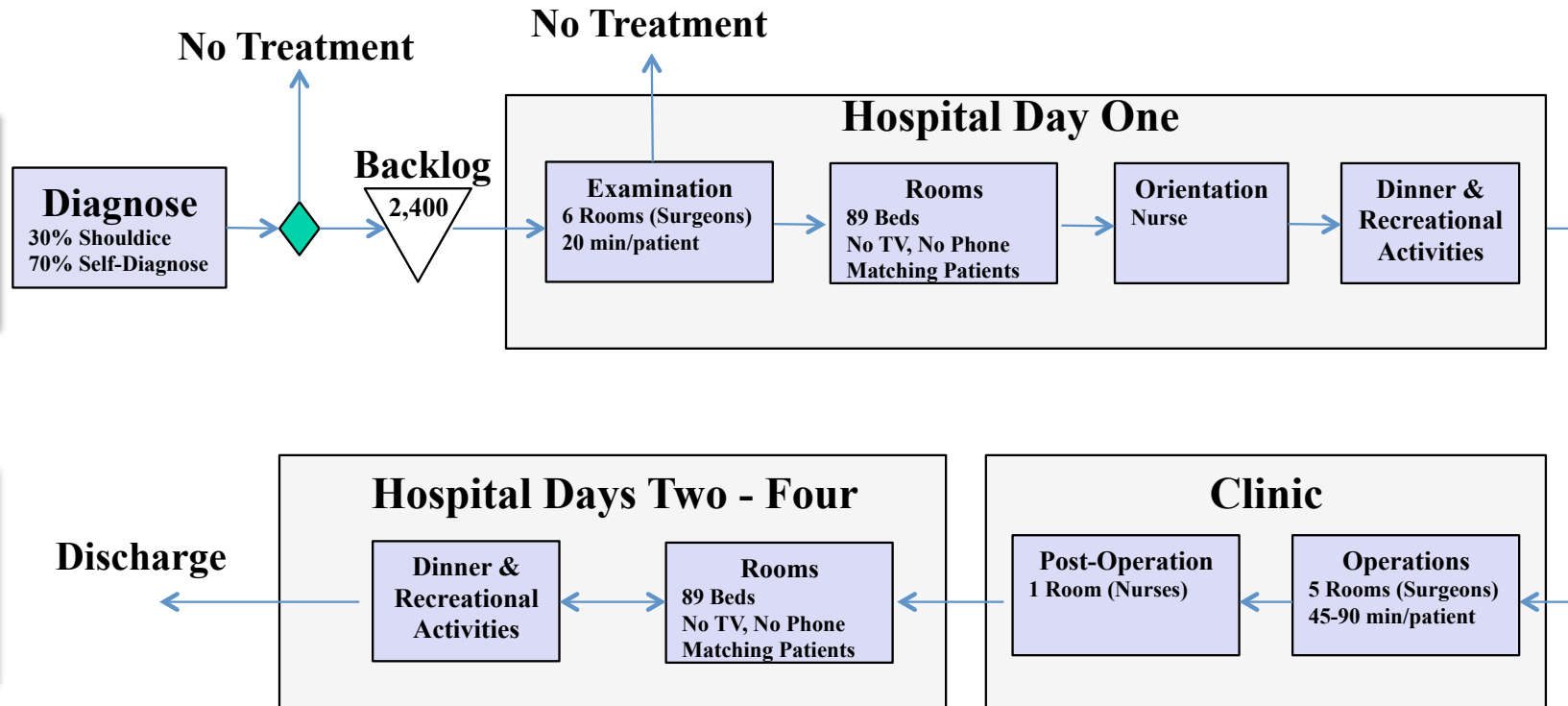
# Business Process: Production Flows



What operational metrics would you use to evaluate the performance of this process?

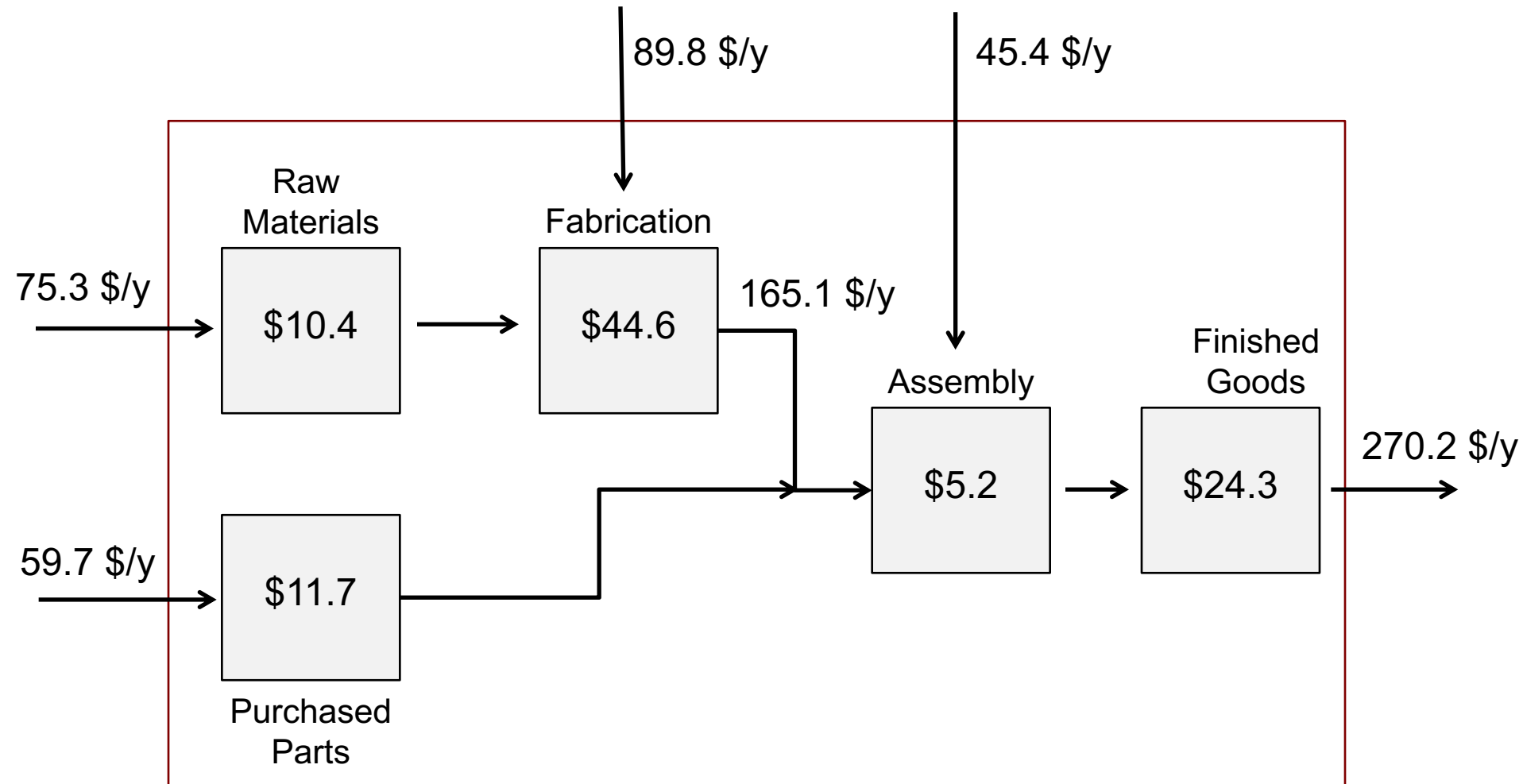


# Business Process: Service Flows



What operational metrics would you use to evaluate the performance of this process?

# Business Process: Cash-to-Cash Flows



# Performance Measures & Capabilities

## What defines a “good process”?

### Financial Measures

- Absolute measures
  - Revenues, costs, operating income, net income
  - Net Present Value (NPV)

- Relative measures
  - ROE
  - ROA, ROI

#### 2012 Financial Performance

(Values in USD Millions)

|             | Walmart | Sears  |
|-------------|---------|--------|
| Sales       | 443,854 | 41,567 |
| COS         | 335,127 | 31,681 |
| Cash        | 6,550   | 754    |
| AR          | 5,937   | 695    |
| Inventories | 40,714  | 8,407  |
| AP          | 36,608  | 2,912  |
| Margin      | 24.5%   | 23.8%  |



# Performance Measures & Capabilities

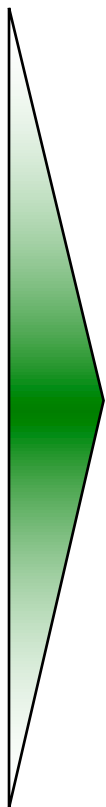
## Operational Capabilities

### ➤ Resource Management Measures

- Production Efficiency
- Resource Utilization
- Flexibility (Volume and Variety)
- Cost Effective Distribution and Logistics

### ➤ Value Creating Measures

- TQM / Lean Manufacturing
- Mass Customization
- Time to Market (Rapid & Reliable Delivery)
- Customer Service
- Supply Chain Management
- Innovation: Products & Processes
- Sustainable & Socially Responsible Ops



“Order Winners”  
based on  
“Capabilities”

# Operations Strategies - Price/Cost

(Dimensions of Customer Value)



## Corporate Strategy

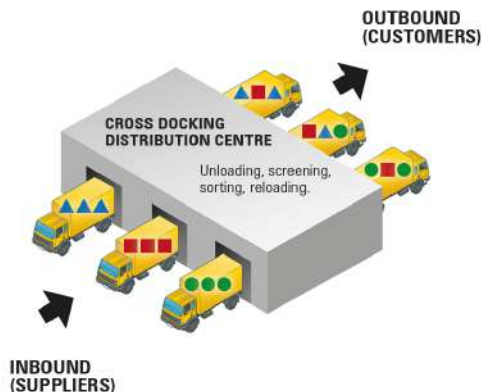
(Gain competitive advantage by) providing customers access to quality goods, when and where needed, at competitive prices.

## Operations Strategy

- Short cycle times
- Low inventory levels

## Operations Structure

- Cross docking
- EDI
- Focused locations
- Communication between retail stores
- Fast transportation system
- RFID (mandate)



# Real-Time Information in Inventory Management

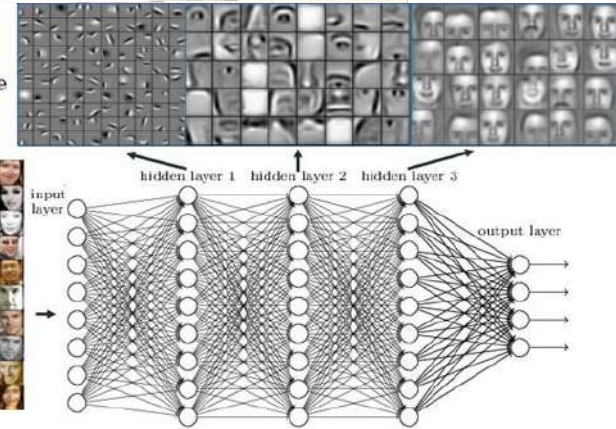
## RF Tags: Keeping the Shelves Stocked

The supply chain of the consumer packaged goods industry works well when confronted by a sudden surge in demand. RF Tags could change what's happening on store shelves. Here's how the system would work.

- 1 A special offer causes Wal-Mart shoppers to snap up boxes of Baby-Dry Pampers.



Deep neural networks learn hierarchical feature representations



- 4 The Wal-Mart IMS is linked to the P&G Supply Chain Management System. Demand spikes reported by RF tags are immediately throughout the
- 5 P&G's logistics system tracks its trucks with locators, and tracks contents with RF readers. Real-time can report





# Operations Strategies - Time

(Dimensions of Customer Value)



“... saving six seconds at a drive-through brings a 1% increase in sales.”

– CEO Jack M. Greenberg

## ZARA



## Fast Fashion

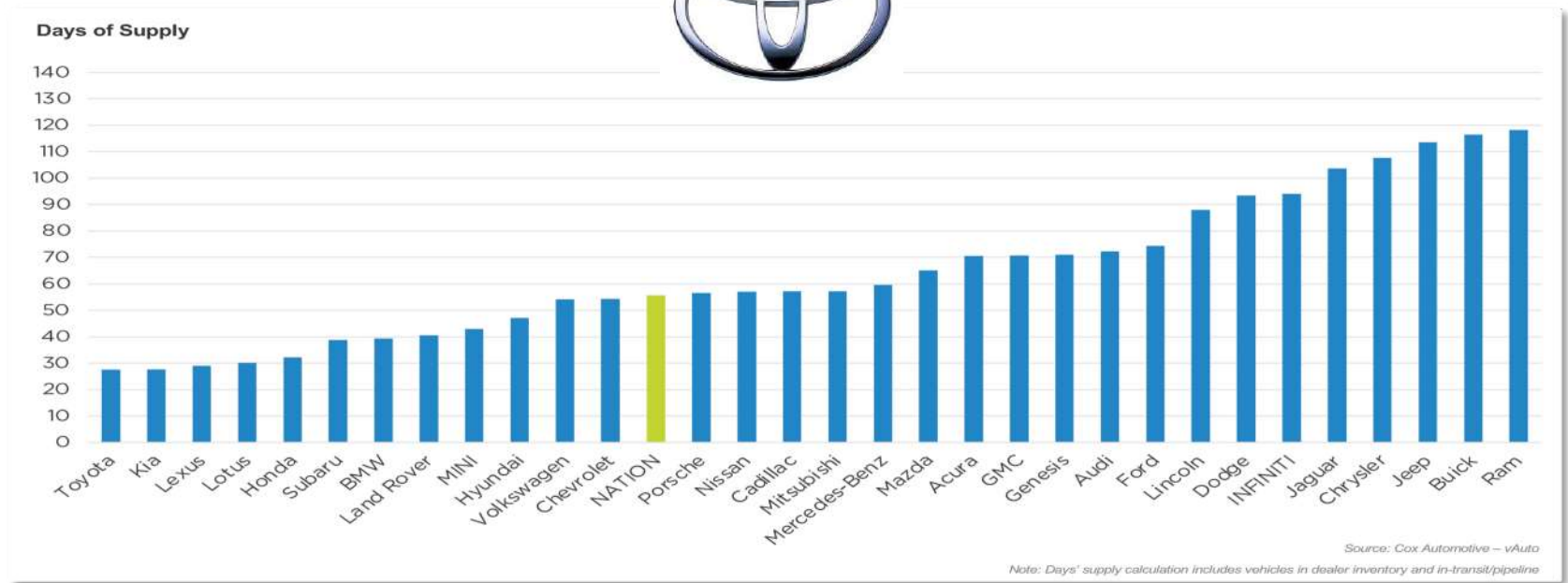
- Sales growth at 20%/yr.
- Net Profit margin at 10%
- NPI once a week,
- 12K new product/year
- 23-30 days of final design to sales cycle.

# Operations Strategies - Quality

(Dimensions of Customer Value)



GUCCI



# Operations Strategies – Variety/Customization

(Dimensions of Customer Value)



## NUMBER OF PRODUCTS ON AMAZON



There are over **600 million products** listed on the marketplace



**12 million** of these items are sold by Amazon themselves



**9 billion+ items** delivered same day or next day by Amazon worldwide in 2024

**More than 60%** of Amazon's sales come from independent sellers



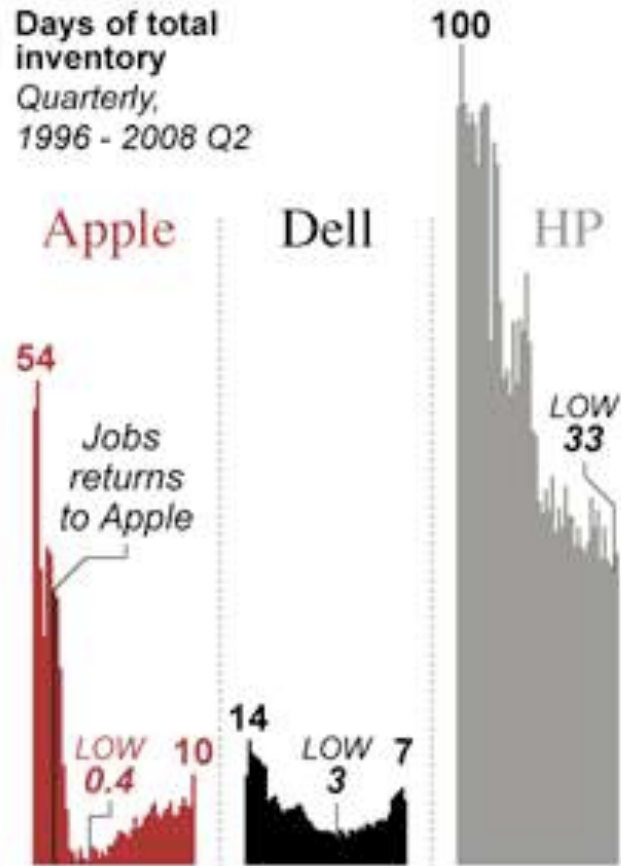
|                  | # SKUs per Store |
|------------------|------------------|
| Avg. Supermarket | 30,000           |
| Walmart          | 150,000          |
| Costco           | 4,000            |
| Home Depot       | 35,000           |



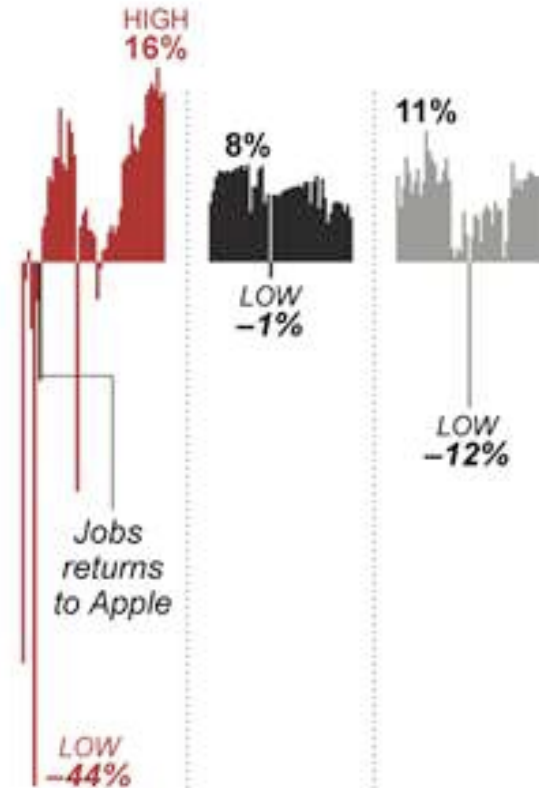
# Apple



Days of total inventory  
Quarterly, 1996 - 2008 Q2



Profit margin  
Quarterly, 1996 - 2008 Q2

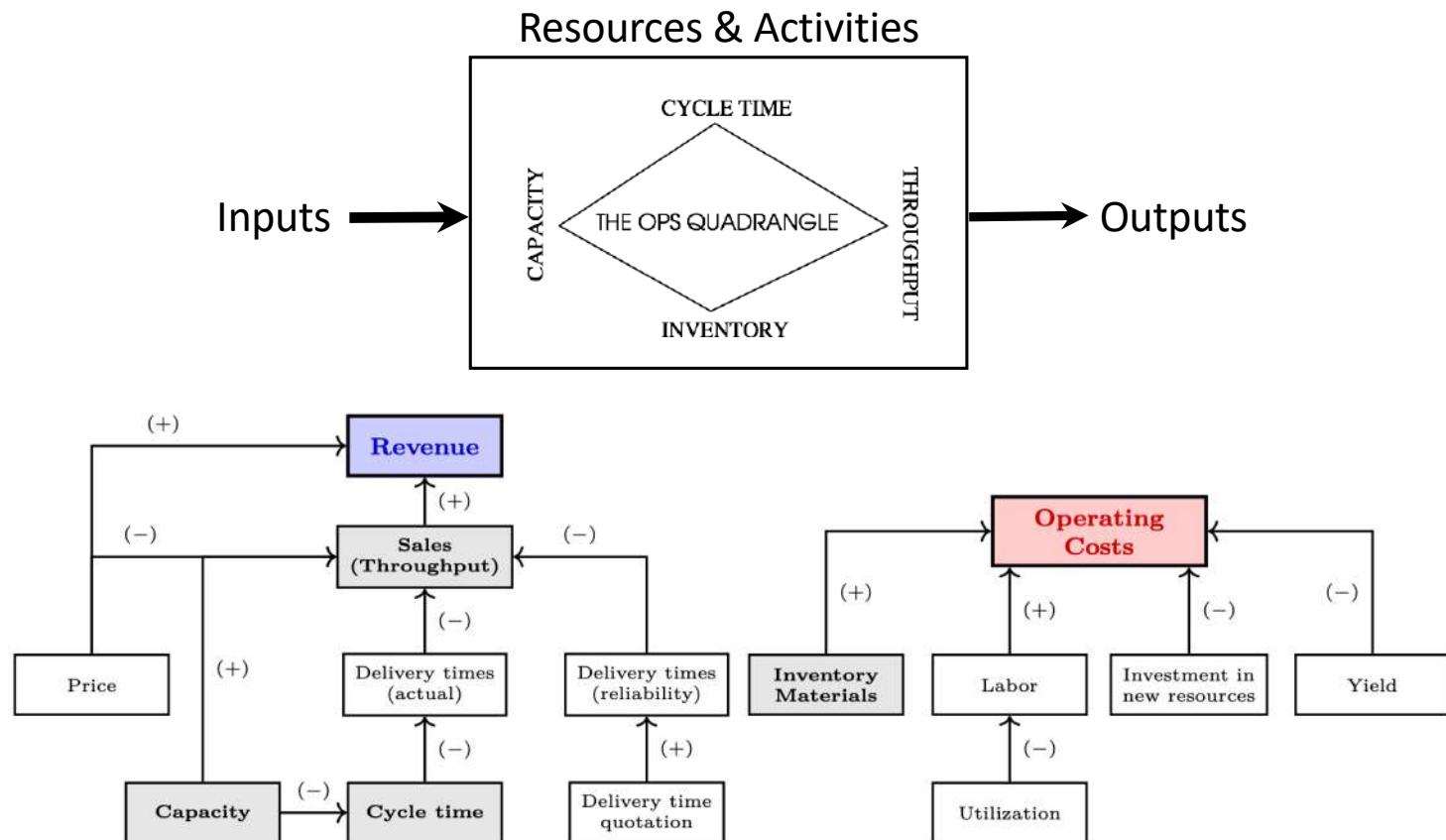


Supply Chain Management [[Link](#), [Link](#)]

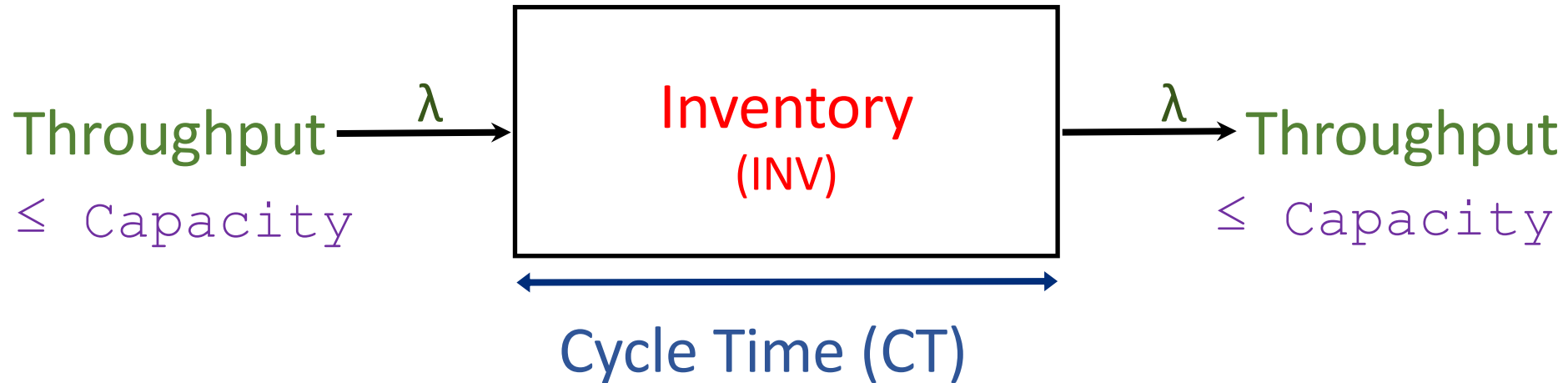
# Business Process Flow Management

*“The best management is a true science, resting upon clearly defined laws, rules, and principles, as a foundation.”*

— Frederick W. Taylor (1856-1915)



# Process Flow Measures



- ♦ **Cycle time:** Time a job spends in a process (time)
- ♦ **Inventory:** Jobs that accumulate in a process (jobs)
- ♦ **Throughput:** Rate at which jobs “come and go” through the process (jobs/time)
- ♦ **Capacity:** Maximum achievable average throughput (jobs/time)

# Bathtub Analogy



## Piling Up

Fannie Mae and Freddie Mac are acquiring foreclosed homes faster than they are selling them, creating a backlog of foreclosed properties that must be sold.

Properties in thousands:

Acquired Sold



# How to calculate?

Suppose  $N$  jobs “come and go” over time duration  $T$ .

$$\text{Average Throughput} = \lambda = N/T$$

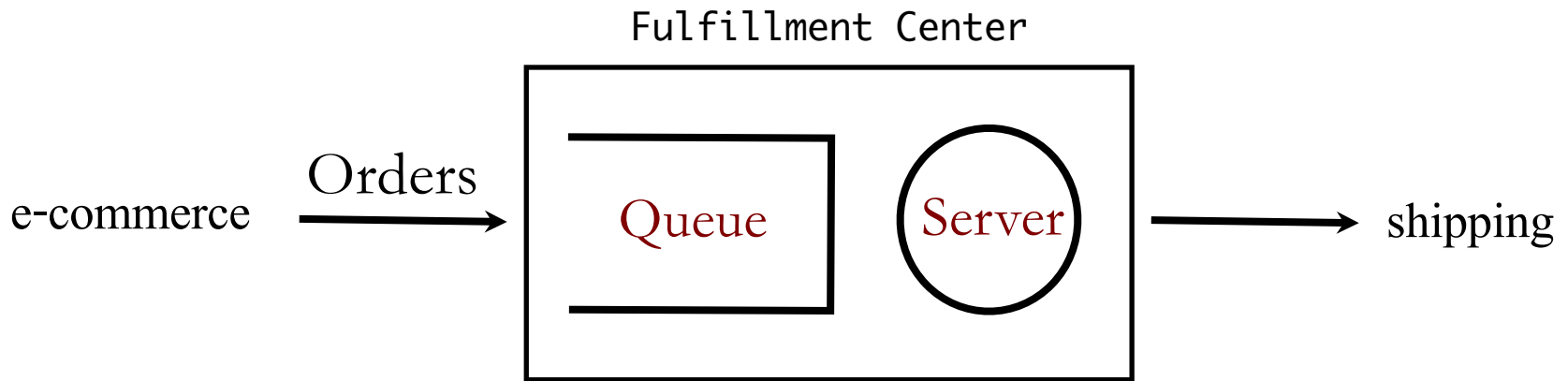
$$\text{Average Cycle time} = \text{CT} = \frac{\sum_{i=1}^N CT_i}{N}$$

where  $CT_i$  is the cycle time of job  $i$ .

$$\text{Average Inventory} = \text{INV} = \frac{\int_0^T \text{INV}(t) dt}{T} = \frac{\text{AREA}}{T}$$



# Example: Packing & Labeling

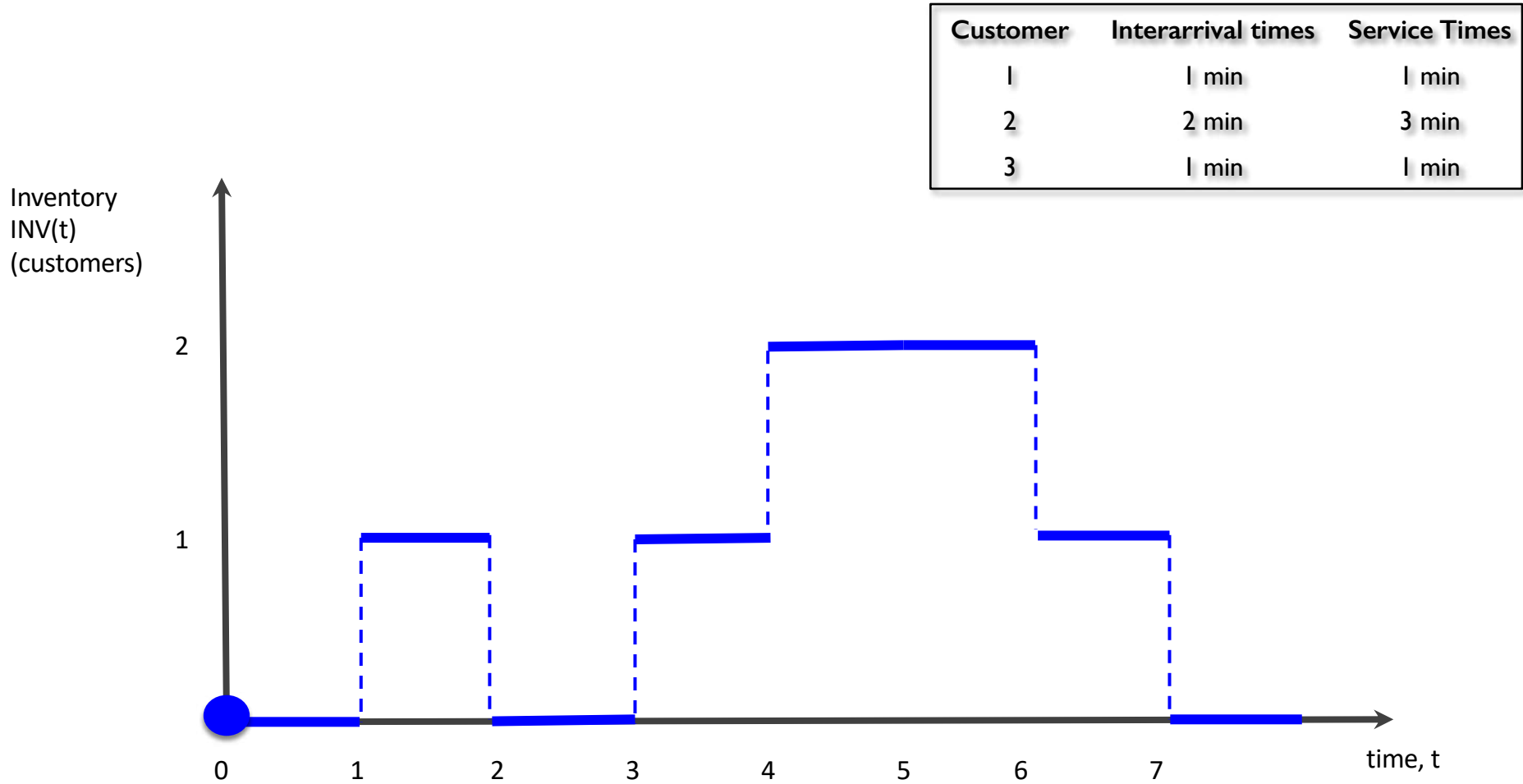


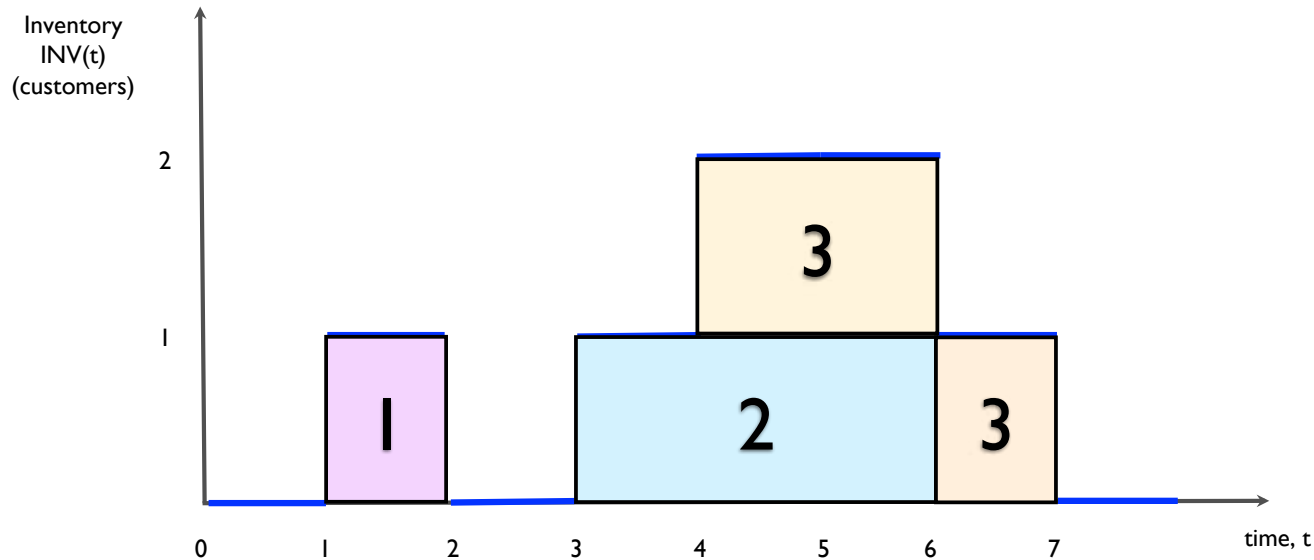
Assume the customers arrive and are served as follows:

| Customer | Interarrival times | Service Times |
|----------|--------------------|---------------|
| 1        | 1 min              | 1 min         |
| 2        | 2 min              | 3 min         |
| 3        | 1 min              | 1 min         |

What are the system's average throughput, cycle time, and inventory?

# Inventory Build-up Chart





$N = 3$  customers

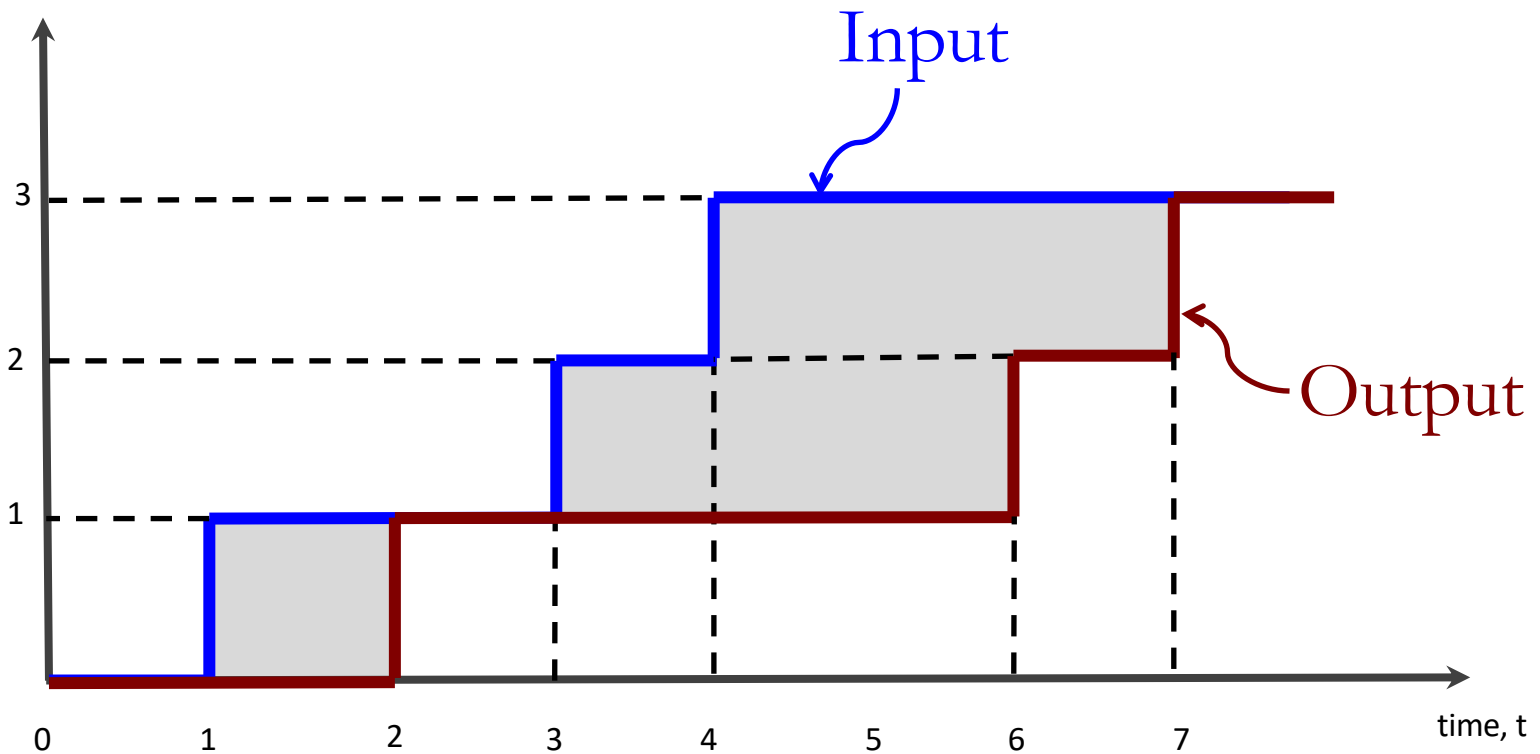
$T = 7$  minutes

**Average Throughput =  $\lambda$  =**  $\frac{N}{T} = \frac{3}{7}$  customers/minute

**Average Cycle time =  $CT$  =**  $\frac{\sum_{i=1}^N CT_i}{N} = \frac{1+3+3}{3} = \frac{7}{3}$  minutes

Any relationship among these numbers?

**Average Inventory =  $INV$  =**  $\frac{\int_0^T INV(t) dt}{T} = \frac{\text{AREA}}{T} = \frac{1+3+3}{7} = 1$  customer



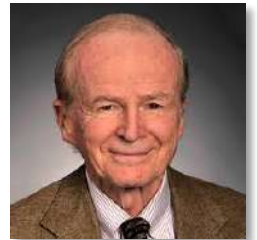
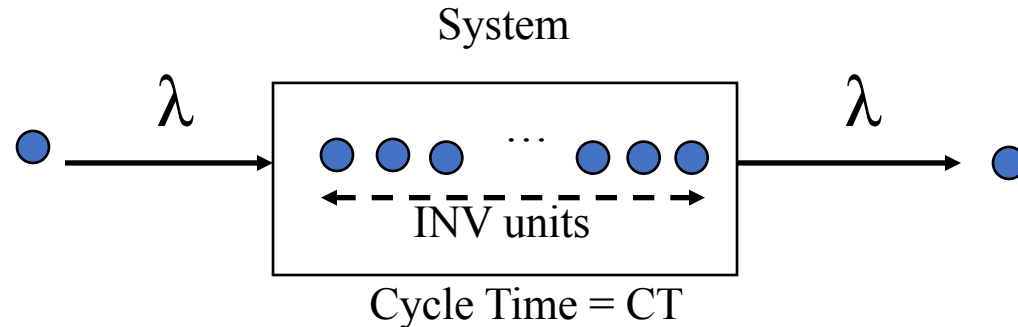
Two interpretations for the area between Input and Output

$$\begin{aligned}
 1) \quad \text{AREA} &= \int_0^T INV(t) dt \Rightarrow \int_0^T INV(t) dt = \sum_{i=1}^N CT_i \Rightarrow \frac{\int_0^T INV(t) dt}{T} = \frac{N}{T} \times \frac{\sum_{i=1}^N CT_i}{N} \\
 2) \quad \text{AREA} &= \sum_{i=1}^N CT_i \\
 &\Rightarrow \boxed{INV = \lambda \times CT}
 \end{aligned}$$

**Remark:** This analysis holds at those times  $T$  at which  $\text{Input}(T) = \text{Output}(T)$ .

Systems that satisfy this conditions are “stable”.

# Fundamental Law of Business Flows



John D.C. Little (1928-2024)

**LITTLE'S LAW:** *the fundamental (long-term) relationship between Inventory (INV), Throughput ( $\lambda$ ) and Cycle Time (T) of a process in steady state is:*

*Inventory = Throughput  $\times$  Cycle Time*

$$INV = \lambda \times CT$$

# Business Flow Examples

➤ **Job Flow:** The Travelers Insurance Company processes 10,000 claims per year. The average processing time is 3 weeks. Assuming 50 weeks in a year, what is the average number of claims “in process”.

➤ **Material Flow:** Wendy’s processes an average of 5,000lb. Of hamburgers per week. The typical inventory of raw meat is 2,500lb. What is the average hamburger’s flow time and Wendy’s turnover?

➤ **Customer Flow:** At London Underground, an escalator at Holborn station serves passengers during the morning rush hour. Under the current policy (stand on the right, walk on the left), the escalator processes an average of 90 passengers per minute. Standing passengers take 30 seconds to ride the escalator, while an average walking passenger takes only 20 seconds. The escalator is 23.1m high, so many passengers choose not to walk, creating congestion and underutilization of capacity.

Managers are considering a new policy in which all commuters stand on both sides of the escalator. Under this policy, the escalator can hold up to 60 passengers at a time. What would be the throughput and the average cycle time under these new operating conditions? What is the effect on the average cycle time of a commuter?





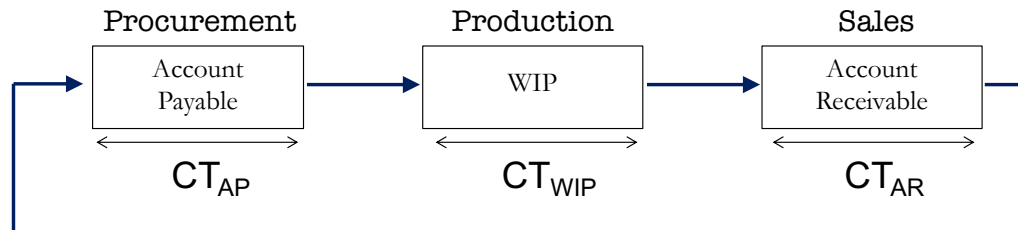
# Business Flows: Cash-to-Cash Cycle

## 2012 Financial Performance

(Values in USD Millions)

|               | Walmart      | Sears        | Target       |
|---------------|--------------|--------------|--------------|
| Sales         | 443,854      | 41,567       | 69,865       |
| COS           | 335,127      | 31,681       | 50,437       |
| Cash          | 6,550        | 754          | 794          |
| AR            | 5,937        | 695          | 6,927        |
| Inventories   | 40,714       | 8,407        | 7,918        |
| AP            | 36,608       | 2,912        | 6,857        |
| <b>Margin</b> | <b>24.5%</b> | <b>23.8%</b> | <b>27.8%</b> |

$$\text{Cash-to-Cash Cycle} = CT_{\text{WIP}} + CT_{\text{AR}} - CT_{\text{AP}}$$



## Flow Time (weeks)

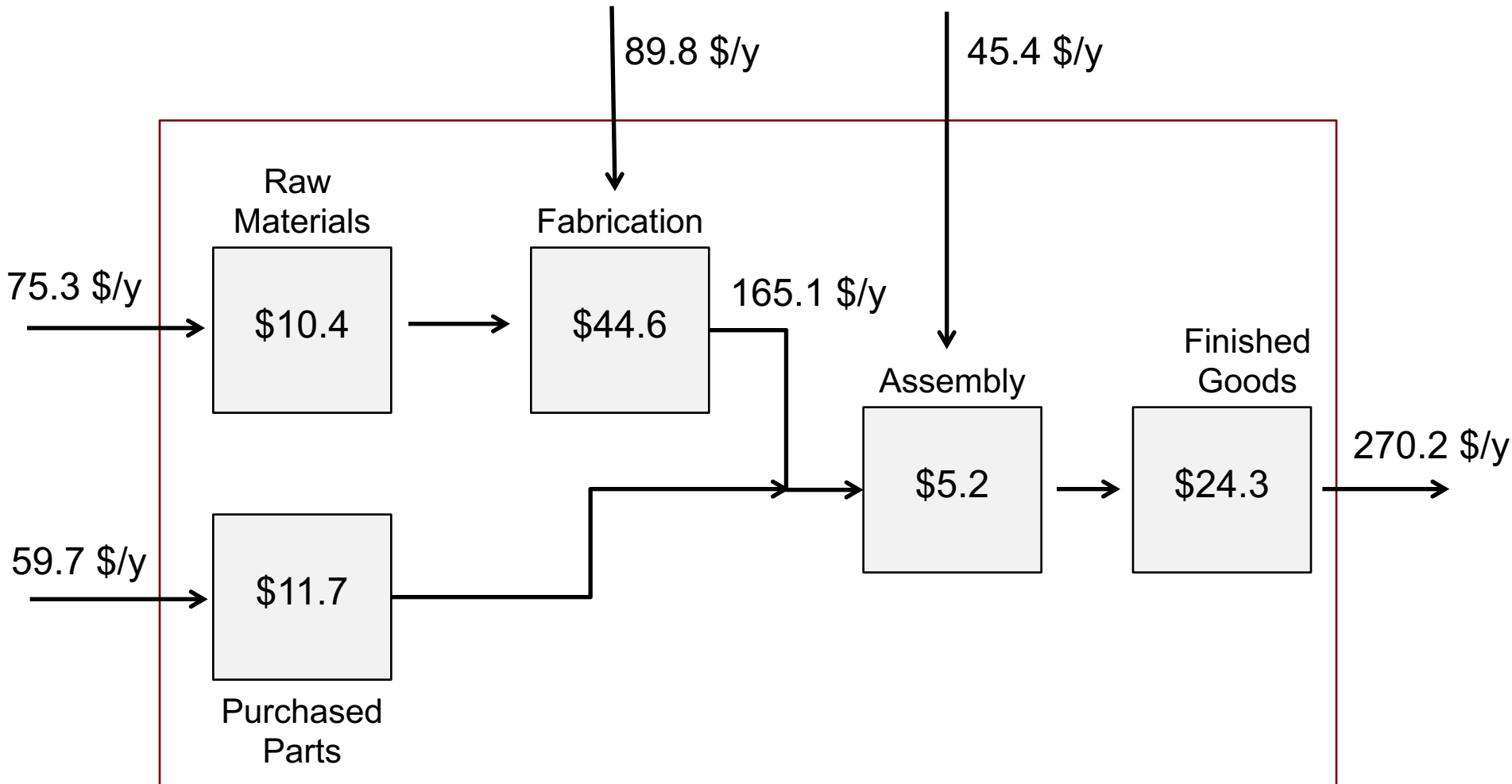
|                     |             |             |             |
|---------------------|-------------|-------------|-------------|
| Cash                | 1.02        | 1.24        | 0.82        |
| AR                  | 0.70        | 0.87        | 5.16        |
| Inventories         | 6.32        | 13.80       | 8.16        |
| AP                  | 5.68        | 4.78        | 7.07        |
| <b>Cash-to-Cash</b> | <b>1.33</b> | <b>9.89</b> | <b>6.25</b> |

# Cash Conversion Cycle (CCC) Benchmarks by Industry

| Industry / Sector       | Typical CCC Benchmark (Days) |
|-------------------------|------------------------------|
| Retail & Consumer Goods | 20 – 40 days                 |
| Manufacturing           | 60 – 90 days                 |
| Technology / Software   | 15 – 30 days                 |
| Healthcare Services     | 45 – 75 days                 |
| Construction            | 90 – 150 days                |
| Agriculture             | 120 – 200 days               |
| Avg. S&P 1500 Companies | 70 - 80 days                 |

Sources: J.P. Morgan Working Capital Index 2024 & Paidnice

# Business Flows : Process Improvement



**Question:** Where would you trim one week of cycle time to free the maximum amount of cash?

# Business Flows : Process Improvement

## Effect of Cycle Time on Balance Sheet

| Item        | Throughput<br>(\$M per week) | Cycle Time<br>(weeks) | Inventory<br>\$M | Target CT<br>(week) | Target Inventory<br>\$M | Inventory Reduction |
|-------------|------------------------------|-----------------------|------------------|---------------------|-------------------------|---------------------|
| RM          | 1.5                          | 6.9                   | 10.4             | 5.4                 | 8.1                     | 2.3                 |
| PP          | 1.2                          | 9.8                   | 11.7             | 5.8                 | 6.96                    | 4.74                |
| FAB         | 3.3                          | 13.5                  | 44.6             | 10.5                | 34.65                   | 9.95                |
| ASSEM       | 5.4                          | 0.9                   | 5.2              | 1.4                 | 7.56                    | -2.36               |
| FG          | 5.4                          | 4.5                   | 24.3             | 4.0                 | 21.6                    | 2.7                 |
|             |                              |                       |                  |                     |                         |                     |
| Payables    | 7.2                          | 5.6                   | 40.1             | 5.0                 | 36.0                    | 4.1                 |
| Receivables | 7.72                         | 7.2                   | 55.7             | 5.2                 | 40.1                    | 15.6                |
|             |                              |                       |                  |                     |                         |                     |
| Total       |                              |                       |                  |                     |                         | 36.986              |

# Key Ideas

- Operations Management is in charge of managing the **transformation** of **Inputs** into **Outputs** by controlling
  - The type of inputs and outputs
  - Flow units
  - A network of activities and buffers
  - Resources and capacities
  - Information structure.
- Effective operations are based on a clear link between business strategy, operations strategy, and operations structure
- What define a “good” operations: **Capabilities**
  - **Cost**: Resource and Cost Efficiency / Productivity
  - **Time**: Time to Market / Mass Customization / Customer Service
  - **Quality**: TQM / Lean Manufacturing
  - **Variety**: Supply Chain Management and Logistics
  - **Innovation**: Products & Processes
- The Process View:
  - Performance Measures and Process Flow Management
  - Bottleneck & Capacity Analysis
  - $INV = \lambda CT$

# After Class Activities

(Assignments are due before class)

## ➤ INDIVIDUAL:

### **TAKEAWAYS #1:**

Contribute one takeaway idea towards your group's possible final project. Do not develop the takeaway at this point, but rather discuss your idea briefly and indicate how it could be developed. (1 paragraph)

## ➤ GROUP:

### **CASE: NATIONAL CRANBERRY COOP**

Throughput Management and Bottleneck Analysis.

### **CASE: MERTON TRUCK COMPANY**

Optimal Resource Allocation